

Random variable HW

1. A random variable X has the following probability distribution.

$X=x_i$	-2	-1	0	1	2	3
$P(X=x_i)$	0.1	K	0.2	2k	0.3	3k

Find i) the value of k ii) Evaluate $P(X < 2)$ and $P(-2 < X < 2)$ iii) Find the cumulative distribution of X and iv) Evaluate the mean and variance of X

2. Determine mean, variance, $E(2X+1)$, $\text{Var}(2X+1)$ of a discrete random variable X given its distribution as follows:

$X=x_i$	1	2	3	4	5	6
$F(X=x_i)$	1/6	2/6	3/6	4/6	5/6	1

3. The probability function of an infinite discrete distribution is given by $P[X = j] = \frac{1}{2^j}, j = 1, 2, 3, \dots, \infty$. Find the mean and variance of the distribution. Also find $P[X \text{ is even}]$, $P[X \geq 5]$ and $P[X \text{ is divisible by 3}]$. Hint: $(1-x)^{-2} = 1 + 2x + 3x^2 + \dots$, $(1-x)^{-3} = \frac{1}{2}[1.2 + 2.3x + 3.4x^2 + \dots]$, $(1-x)^{-1} = 1 + x + x^2 + \dots$.
4. The cumulative distribution function (cdf) of a random variable X is $F(x) = 1 - (1+x)e^{-x}, x > 0$. Find the probability density function of X, mean and variance of X.
5. If the density function of a continuous random variable X is given by

$$f[x] = \begin{cases} ax & , 0 \leq x \leq 1 \\ a & , 1 \leq x \leq 2 \\ 3a - ax & , 2 \leq x \leq 3 \\ 0 & , \text{otherwise} \end{cases}$$

1) find the value of a. 2) find the cumulative

distribution function of X. 3) let A and B be event of getting between 1 and 2 and greater than 1.5. 4) Are A and B independent events? Hint: For independent check $P(A \cap B) = P(A) \cdot P(B)$.